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Computer Science & Engineering

R23 B.TECH CSE IV YEAR I SEM :: AC Year:2026-2027
MAJOR PROJECT PHASE-1 – ABSTRACT FORM

PROJECT TITLE

AI CodeSense: Intelligent Code Review and Documentation Assistant

PROBLEM STATEMENT:

Manual code reviews, bug detection, security analysis, and documentation are time-consuming and prone to errors. Using multiple independent tools makes the development process inefficient and increases the risk of overlooked issues.

STUDENT DETAILS

S.No	Roll No.	Student Name	Signatures
1	2311CS010234	GORLA NITHESH REDDY	
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ABSTRACT:

Software development requires continuous code review, bug detection, security analysis, and technical documentation to maintain software quality and reliability. However, manual code reviews are time-consuming, documentation is often incomplete or outdated, and developers frequently rely on multiple independent tools for code analysis, security scanning, and documentation generation. This fragmented workflow reduces development efficiency and increases the likelihood of overlooked issues and inconsistent documentation.

To address these challenges, this project proposes **AI CodeSense: An AI-Powered Code Review and Technical Documentation Platform**, an intelligent web application that assists developers throughout the software development lifecycle. The platform enables users to upload source code, paste code snippets, or connect GitHub repositories for automated analysis. Artificial Intelligence is utilized to detect bugs, identify code smells, analyze code complexity, recommend performance improvements, generate technical documentation, create UML descriptions, and provide intelligent code explanations through an AI-powered chatbot.

By integrating automated code analysis with AI-powered documentation and developer assistance, AI CodeSense provides a unified platform that improves software quality, accelerates the code review process, enhances documentation consistency, and increases overall developer productivity. The system is designed to reduce the effort involved in manual code inspection and documentation while providing developers with actionable insights to improve their code.

TECHNOLOGIES / TOOLS USED:

Frontend: React.js

Backend: Spring Boot

AI/Analysis Service: Python, FastAPI

Database: MongoDB

AI API: Google Gemini API

Repository Integration: GitHub

Version Control: Git

Project Coordinator **Project Guide**